

Nodes on the Fabric of interest in the local network. Moreover, Known Answer Suppression [RFC 6762] SHOULD be employed in such cases to further minimize the number of unnecessary responses to such a query.

3. When resolving the operational service record of another Node, a Node SHOULD use an SRV query for the desired operational service instance rather than doing general enumeration of all nodes (e.g. PTR query) followed by filtering for the desired service instance. This recommendation reduces the amount of multicast traffic generated on-link when Multicast DNS is used, and reduces latency to successful service resolution.
4. Since proxied DNS-SD service discovery MAY be in use within a given network, and service record caching is expected of DNS-SD clients, Nodes SHOULD NOT use DNS-SD as an operational liveness determination method. This is because there may be stale records not yet expired after a Node becomes unreachable which may still be available.

#### 4.3.2.7. Operational Discovery DNS-SD Examples

The example below simulates a commissioned Matter Node advertising its availability for control via the Matter protocol.

Examples are shown using both the **dns-sd** command-line test tool and the **avahi** command-line test tool. The **dns-sd** command-line test tool is included in all versions of macOS. It is installed as a DOS command with Bonjour for Windows, and is available on Linux by installing the **mDNSResponder package** [https://github.com/balaji-reddy/mDNSResponder]. The **avahi** command line-test tool is available from the **Avahi project** [https://github.com/lathiat/avahi] for most Linux distributions.

This example is given for illustrative purposes only. Real Matter Nodes and controllers would not use a command-line test tool for advertising and discovery. Real Matter Nodes and controllers would use the appropriate DNS-SD APIs in their respective chosen programming languages.

Consider a device on Wi-Fi using the 48-bit device MAC address of B75AFB458ECD as its host name. DNS-SD records for can be set up as follows:

```
dns-sd -R 87E1B004E235A130-8FC7772401CD0696 _matter._tcp . 22222
```

or

```
avahi-publish-service 87E1B004E235A130-8FC7772401CD0696 _matter._tcp 22222
```

The port number 22222 is given here purely as an example. One of the benefits of using DNS-SD is that services are not constrained to use a single predetermined well-known port. This means that multiple instances of the Matter Node control service can run on the same device at the same time, listening on different ports [RFC 6760]. The port, along with the IPv6 address, is discovered by the Matter controller at run time.

Avahi only sends a single AAAA record. To force the link-local address to be used, use **avahi-publish-address**. For example: